Change Detection Analysis Report

1. Introduction

This report demonstrates the process of performing change detection using multi-sensor Earth Observation (EO) data. The focus is on integrating Sentinel-1 (SAR) and Sentinel-2 (Optical) data for a selected area of interest, using Python for automation, analysis, and visualization.

2. Objectives

- Demonstrate ability to programmatically find and ingest EO data (API or Python clients).
- Perform pre-processing and coregistration of optical and SAR imagery.
- Conduct a change detection analysis using both data sources.
- Present results clearly with reproducible code, report, and visual outputs.

Data sets used and Preprocessing

1. Sentinel-2 Level-2A imagery was accessed via the Copernicus Open Access Hub API of 2017-02-10 and 2019-02-10. Since L2A products are radiometrically and atmospherically corrected, no additional preprocessing was performed. The focus was on downloading specific bands and clipping to the region of interest (ROI) for multispectral analysis and classification. Efficient, cloud-based access ensured reproducibility.
2. Sentinel-1 GRD products were fetched using the Copernicus Open Access Hub API of 2017-01-29 and 2019-02-12, descending orbit, IW swath. Preprocessing steps included precise orbit correction, thermal noise removal, radiometric calibration to sigma0, speckle filtering, and terrain correction. The processed image was clipped to the ROI. These steps improve geolocation accuracy, reduce noise, and ensure the SAR image aligns with optical data for meaningful fusion and analysis.
3. The four spectral bands (B2, B3, B4, B8) from Sentinel-2 were stacked and spatially aligned. Both images were clipped to the same ROI for spectral comparability. The resulting datasets include optical reflectance and SAR backscatter, offering complementary information for land cover classification and change mapping across different environmental and seasonal conditions.

Sentinel-1 API (Radar Imaging)

The Sentinel-1 API provides access to synthetic aperture radar (SAR) imagery through three main endpoints. Authentication uses OAuth2 password grant at identity.dataspace.copernicus.eu to obtain bearer tokens. Catalog search via catalogue.dataspace.copernicus.eu/odata/v1/Products filters products using OData queries with parameters: Collection ('SENTINEL-1'), geographic boundaries (WKT POLYGON), date ranges (ISO 8601 format), product types (GRD/GRD-HD/SLC), and orbit direction (ASCENDING/DESCENDING). Download service at zipper.dataspace.copernicus.eu retrieves ZIP archives containing SAR data with VV/VH polarizations at 10m resolution, ideal for all-weather land monitoring.

Sentinel-2 API (Optical Imaging)

The Sentinel-2 API delivers multispectral optical imagery through identical endpoint structure. Authentication remains consistent with Sentinel-1. Catalog search filters Sentinel-2 products using Collection ('SENTINEL-2'), geographic intersections, temporal ranges, processing levels (L1C for top-of-atmosphere, L2A for atmospherically corrected), and critically, cloud cover percentage filters. Products contain 13 spectral bands at 10m/20m/60m resolutions spanning visible to SWIR wavelengths. Downloads provide complete .SAFE archives with JP2 band files, metadata, and quality masks. The API enables vegetation monitoring, land classification, and environmental analysis requiring clear-sky optical observations.

Methods used:

Gram-Schmidt Fusion (2017 and 2019)

Gram-Schmidt fusion combines high-resolution panchromatic or SAR data with multispectral bands. It enhances spatial resolution while preserving spectral characteristics. Benefits include improved visualization and feature discrimination. However, it may introduce spectral distortion or artifacts if calibration isn't precise. No training data required. It creates two fused images for cluster-based change detection.

Pros:

- Enhances spatial resolution and visual clarity
- Preserves spectral characteristics reasonably well
- Suitable for multi-sensor fusion
- Easy to implement and reproduce

Cons:

- Can introduce spectral distortion
- Requires accurate geometric alignment
- Sensitive to noise in SAR data
- Less effective without good radiometric calibration

K-Mean Clustering Method:

K-Means clustering was applied to both fused images for 2017 and 2019, using 5 clusters corresponding to key land cover classes. This unsupervised method requires no labeled training data. Transition analysis indicated significant conversion from barren to agricultural, with minimal urban expansion. The approach revealed meaningful change dynamics without extensive manual annotation.

Pros:

- Fast and easy to implement
- No ground truth required
- Useful for preliminary or large-scale analysis
- Handles large datasets

Cons:

- Cluster labels may shift between dates
- Noise-sensitive; may misclassify mixed pixels
- No spatial context (pixel-based)
- Requires cluster count preselection

False Positives/Negatives

- **False Positives:** Mixed pixels around water bodies or edges, speckle misinterpreted as change.
- **False Negatives:** Gradual or spectral-only changes undetected by unsupervised methods.
- Validation was limited due to a lack of ground truth. Some clusters were misclassified when features had similar reflectance.

Recommendations

- **Workflow Improvements:** Incorporate supervised classification using ground truth or high-res labels for better accuracy.
- **UI Suggestions:** Build a QGIS plugin or web interface for automated fusion, clustering, and visual output generation.

Result:

Fusion Output: Sentinel 1 and Sentinel 2 Fusion Result of 2017 & 2019

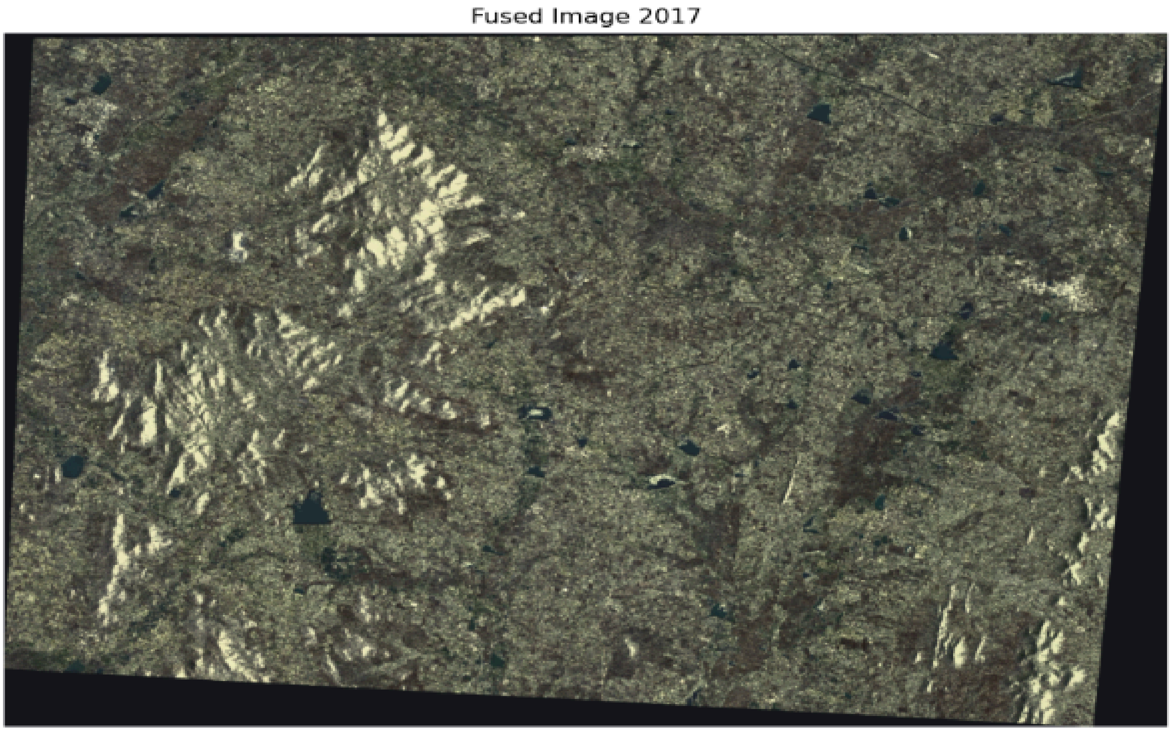


Fig.1 Fused image of Sentinel 1 and 2 2017

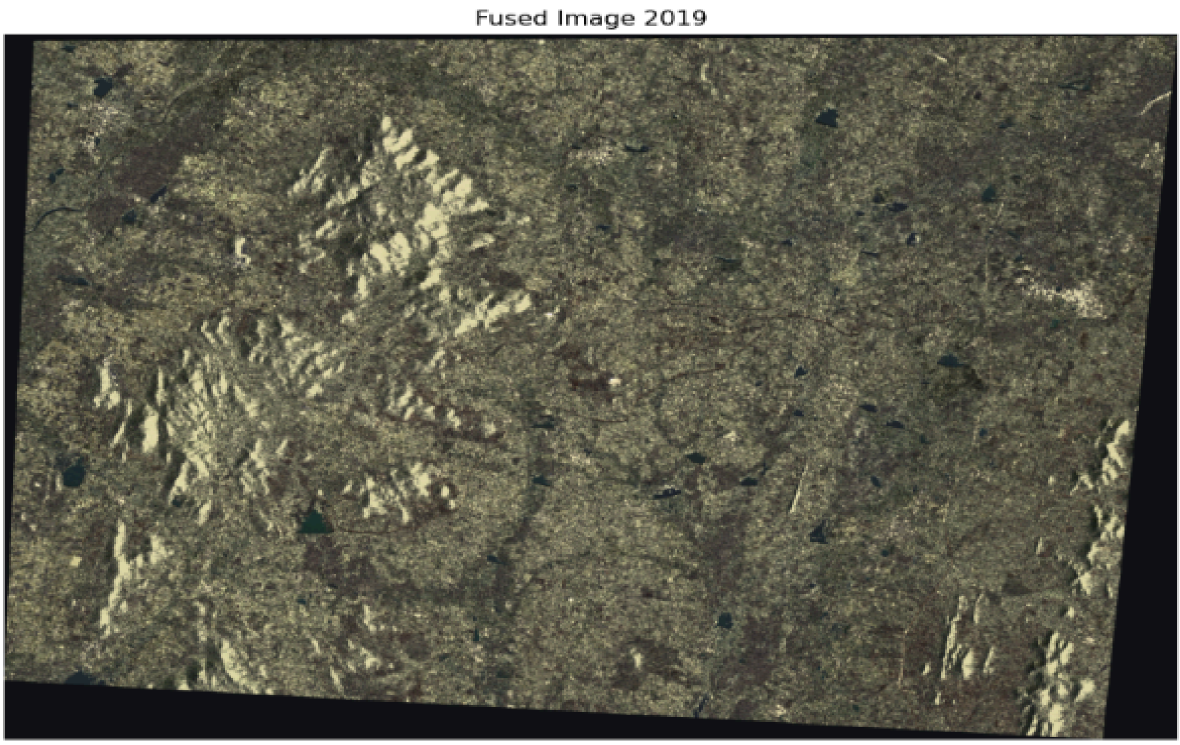
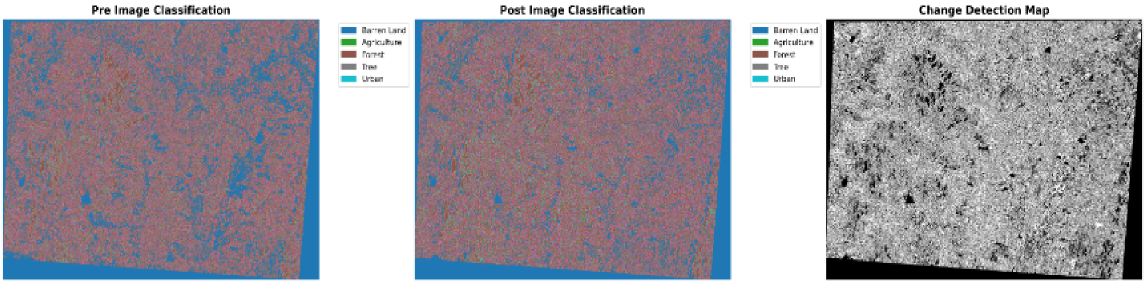


Fig.2 Fused image of Sentinel 1 and 2 2019

Outputs:



<Figure size 640x480 with 0 Axes>

Fig.3 Pre and Post Image Classification

The above figure gives you the information about the pre and post-image classification of the years 2017 and 2019

Fig.4 Forest Area of Vijayawada

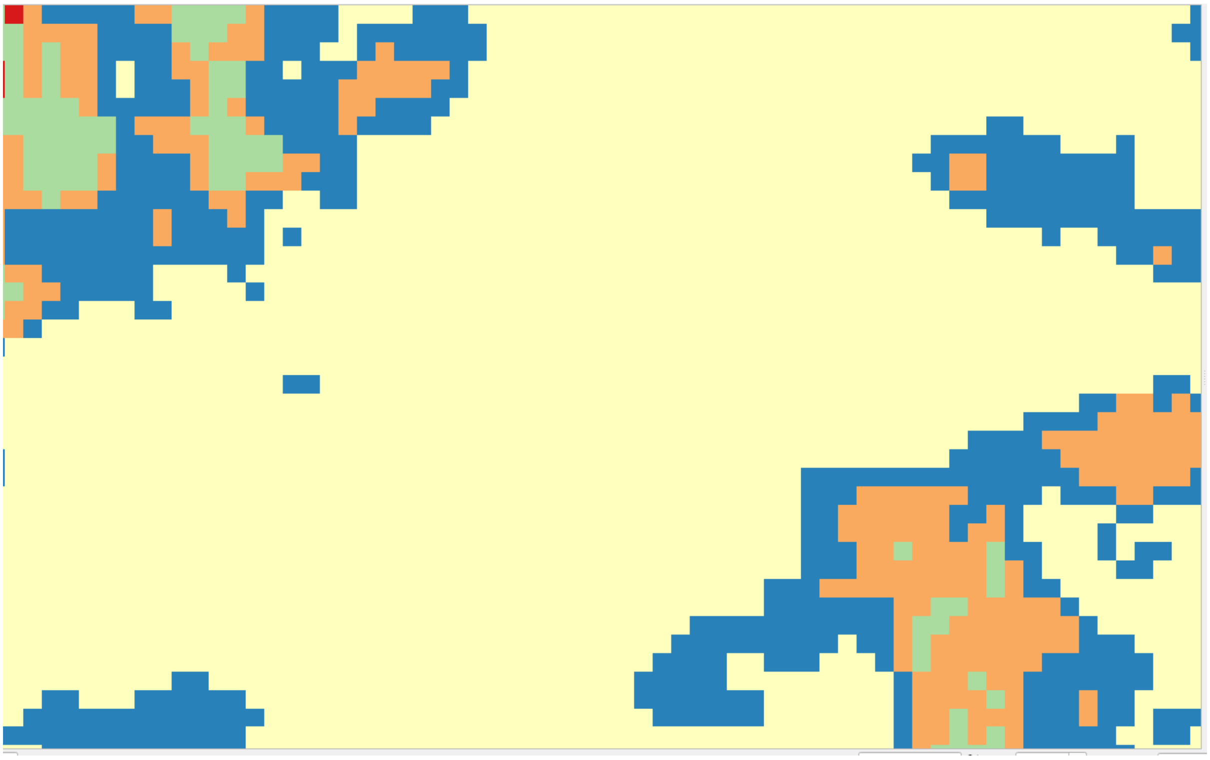


Fig.5 Pre-Image Classification of Forest area in Vijayawada

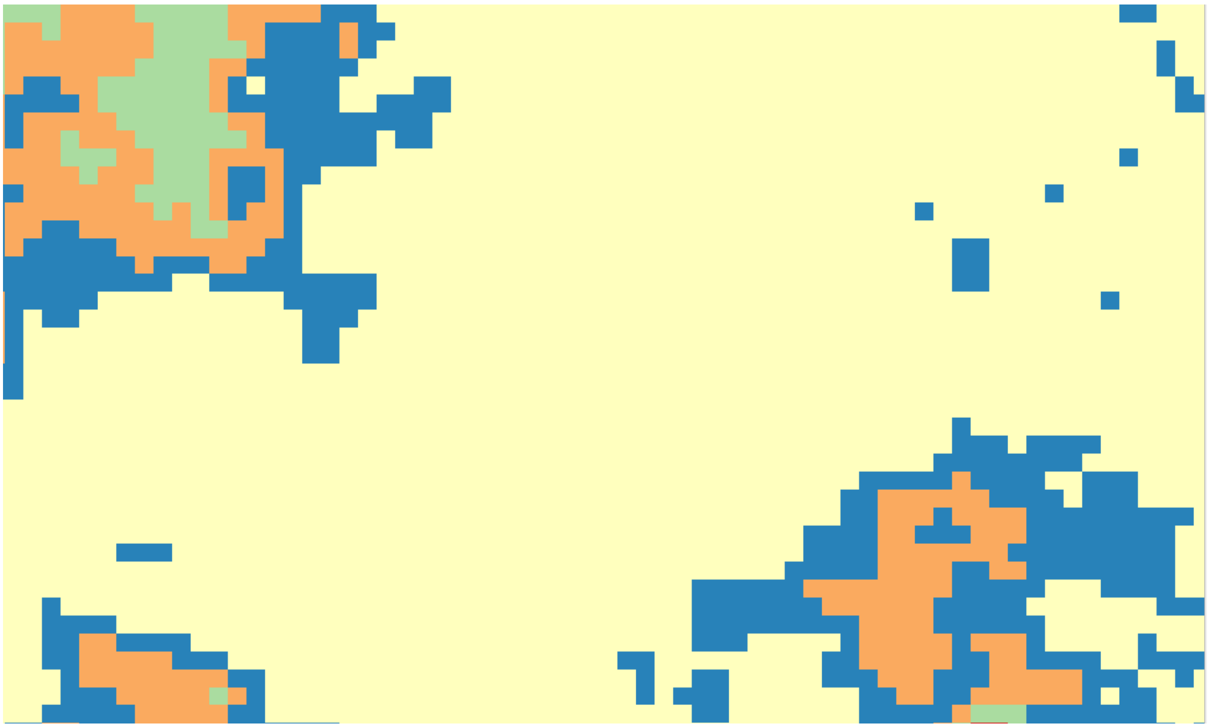


Fig.6 Post-Image Classification of Forest area in Vijayawada

Here in the above figures you can see the pre and post image classification of forest in yellow colour how it has changed from 2017 to 2019.

Fig.7 Barren land in Vijayawada

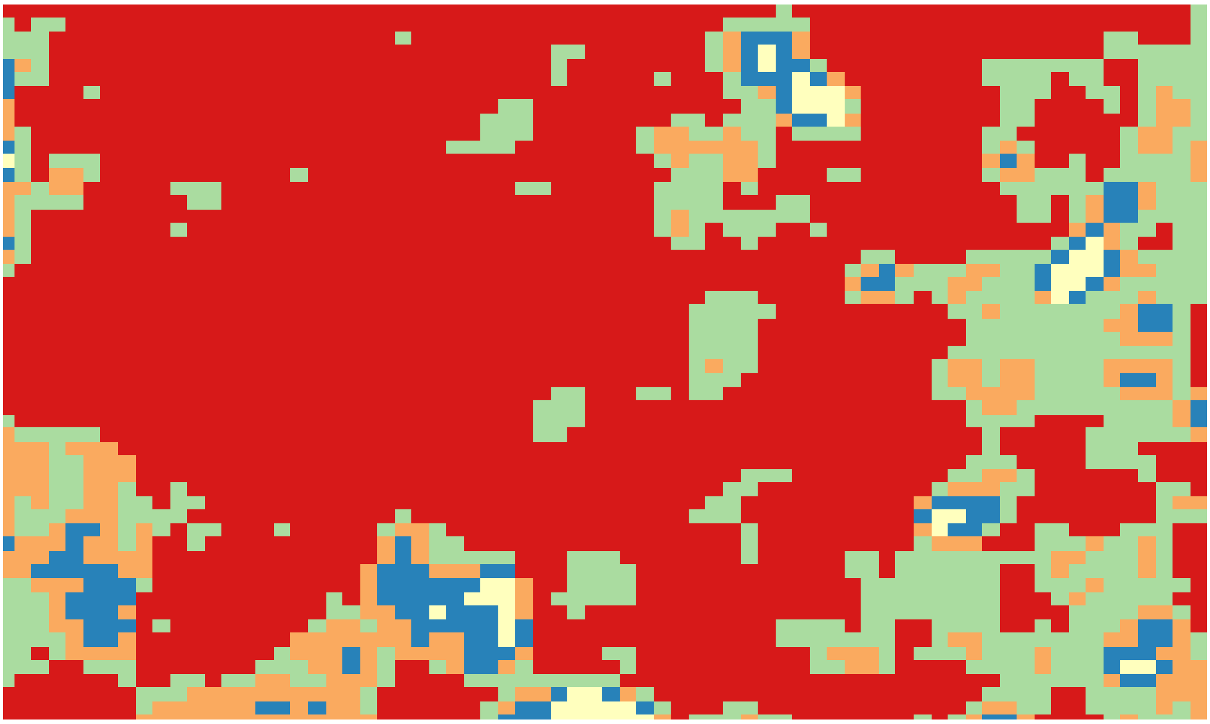


Fig.8 Pre-Image Classification of Barrenland in Vijayawada

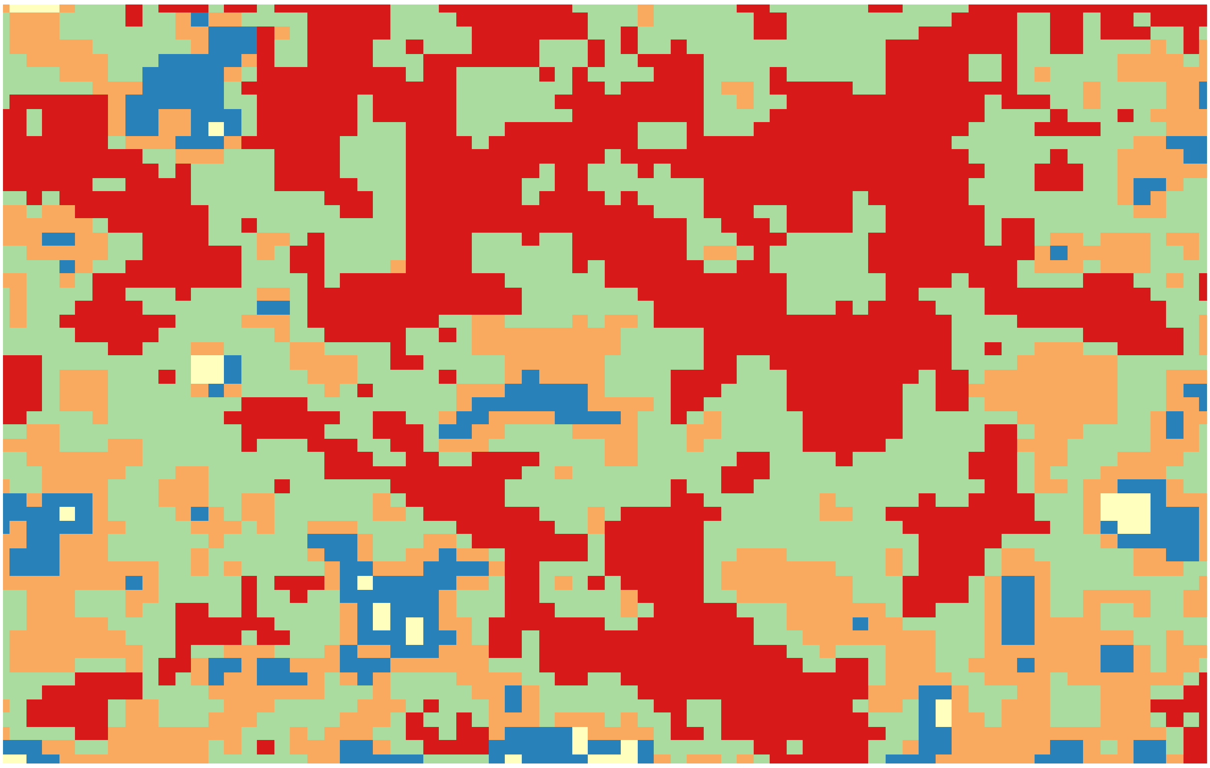


Fig.9 Post-Image Classification of Barrenland in Vijayawada

Here Red color indicates the Barren land but it is converted to green in most of the pixels as you see in the pre and post image classification as shown in Fig.8 and 9.

Conclusion

The change detection results showed a notable shift from barren land to agricultural use, indicating either expansion of farming activities or cultivation of previously unused land. This land cover transition may signal economic development, policy-driven agricultural intensification, or seasonal crop cycles. On the other hand, urban areas exhibited very little change between 2017 and 2019, implying controlled urban growth or infrastructure stability. The use of an unsupervised clustering approach like K-Means enabled these insights without requiring labeled training data or extensive manual mapping, enhancing automation in land change monitoring.

Githublink:

https://github.com/sharath1k/Fusion_Change_Detection_2017-to-2019